



MASTERMIND
INDIA

Spirit of India Special

THERMAL DYNAMICS

By Nandini Vaish

It was when living in Boston that Kranthi Kiran Vistakula faced a too-hot-too-cold problem. A student of mechanical engineering at the Massachusetts Institute of Technology, Vistakula would swathe himself in multiple layers of clothing in winter, most of which he would have to take off the minute he stepped indoors. Irritated, he thought, "Why can't I just make do with one thing?"

In 2006, he went about creating a temperature-controlled jacket that could keep him both warm in winter as well as cool in summer. He started his company, Dhama Apparel Innovations, with an initial investment of \$10,000 from his own savings and from the scholarship he had been awarded.

The sleeveless jacket, called ClimaGear, cools the torso, as it maintains the core body temperature. The jacket has plastic discs which work on the Peltier Effect (named after the French physicist Jean Charles Peltier) or the principle of thermo electricity. Each disc, called a peltier, converts electrical energy into heat and is connected to a bigger, plastic tile sewn into the garment. This means that junctions of dissimilar metals are heated or cooled, depending upon the direction in which electric current is passed through them. So heat generated by current flowing in one direction is absorbed if the current is reversed. Vistakula used a lithium polymer laptop battery to

power his product as it was easily available and also had the highest energy density among the commonly available batteries.

The challenge, however, was to make it wearable. The first prototype he prepared weighed 3 kg. The second was down to 2 kg, but still too heavy. The final product weighs around 900 gm (250 gm of which is the battery), less than a winter coat which typically weighs 1.5 kg. It can maintain temperature between 18°C and 40°C. It lasts three to four hours on one charge, while the overall life of the jacket is about three years or 1,000 charges. Military personnel, outdoor workers and sports enthusiasts, who often operate in harsh environments, may find this jacket useful.

In 2008, Vistakula was awarded a government grant of Rs 12 lakh. The National Institute of Design (NID), Ahmedabad, which was involved in the selection process, offered to incubate his company, which is housed there. He has six employees and four interns who are now working actively towards business development. He also got venture capital funding from Reliance Venture and Mumbai Angels, who helped bring in networking and contacts apart from the funding. A few months ago, they diversified into other products like shoes, scarves, helmets and medical pads. ClimaNeck is a neck accessory, and ClimaShoe is a heating/cooling shoe while ClimaMet is a helmet. All allow the wearer to maintain the body at a specified

Kranthi
KIRAN VISTAKULA, 30

INNOVATION

Temperature-controlled apparel

COST

The jacket costs Rs 40,000. Other products range from Rs 4,000-7,000.

WHAT IT DOES

Allows the wearer to maintain a specified body temperature regardless of the external environment

BASED ON THE PELTIER EFFECT, THE CLIMAGEAR CAN BE ESPECIALLY USEFUL FOR MILITARY PERSONNEL AND SPORTS ENTHUSIASTS WHO OFTEN OPERATE IN HARSH ENVIRONMENTS.

VISTAKULA WITH THE CLIMAGEAR JACKET